MEDICATION FORMULARY:
CANADIAN FORCES AEROMEDICAL EVACUATION FLIGHT REVISION

NOTE: This formulary is applicable to all CF AE activity, including Strategic, Forward, CSS, and SAR capabilities. Specific Drug Lists for each AE activity are to be drawn from this formulary.

Authorized for use by the 1 Canadian Air Division Surgeon per electronic signature below.

Version 1.2

Approval for Use:

GRAINGER, RICHARD 433 Digitally signed by GRAINGER, RICHARD 433 Date: 2021.09.19 19:42:50 -04'00'

Version History

Version No.	Description of Change	Date
1.0	Initial Issue	11 Sept 2021
1.1	Addition of Ertapenem, Revision to TXA, Fc	16 Sept 2021
1.2	Addition of Amiodarone, D10W, D50W, Hydroxycobalamin, Insulin (IVP), Metoprolol, Ondansetron, Sodium Bicarbonate Revision to IVP Dimenhydrinate, Diphenhydramine, Epinephrine, Fentanyl, Haldol, Ketamine, MgSO4 Revision to IV Infusion Epinephrine, Insulin, TXA	19 Sept 2021

PUSH MEDICATIONS: CF AE Flt Revision

MEDICATION	INDICATION	FORMULATION	FINAL CONC	USUAL DOSING	NOTES
Amiodarone	-Cardiac Arrest -Antiarrhythmic	-Administer undiluted	-150mg/3mL	-Arrest: 300mg x 1 then 150mg x 1 -Antiarrhythmic: 150mg over 10 mins	-Caution hypotension
Atropine	-Symptomatic bradycardia -Excessive secretions (i.e.Ketamine)	-Amp of 1mg/mL (draw up in 1 or 3mL syringe- do not dilute) OR -Vial of 0.4mg/mL (draw up 5mL atropine + add 5mL NS) OR -Pre-filled syringe of 1mg/10mL	-1mg/mL OR -0.2mg/mL OR -1mg/10mL	-0.5mg (0.5mL) q 3-5 min OR -0.5mg (2.5mL)	**Caution: different concentrations available** -Max 3mg total -10-20mcg/kg for Ketamine -Nerve Gas Exposure: 2-6mg IV/IM q 5-10min titrate to bradycardia & bronchorrhea/bronchospasm
Calcium Chloride **($CaCl_2$)**	-Massive transfusion -Hypocalcemia-Ca2+ <7.0mg/dL Ionized Ca2+ <0.8mg/dL -Hyperkalemia protocol	-1g 10% $CaCl_2$ (pre-filled syringe)	-100mg/mL -May be given in minibag (1 ampule CaCl 10% provides 27.2mg/mL elemental calcium)	-500-1000mg (5-10mL) over 2-5min -1g for every 4 units of PRBCs given (d/t citrate)	-**IVP: RN or MD only** -Med Techs: must give in 100mL minibag -Central line / large bore IV only
Calcium Gluconate **(Ca-Gluc)**	-Massive transfusion -Hypocalcemia-Ca2+ <7.0mg/dL Ionized Ca2+ <0.8mg/dL -Hyperkalemia protocol	-1g 10% Ca-Gluc	-100mg/mL (1 ampule CaGluc 10% provides 8.9mg/mL elemental calcium)	-1g over 2-5min	-1g for every 4 units of PRBCs given (d/t citrate)
Dexamethasone	-Corticosteroid	-Doses up to 10mg IV	-Vial: 20mg in 5mL -Administer undiluted, or dilute dose in a 10mL NS -May be given in minibag	-1-10mg over 2-5min	-Larger doses are used in septic shock, cerebral edema, bacterial meningitis -Max dose: 80mg/24h -IV dose given rapidly in an awake pt will cause intense perineal burning/itching
D10W	-Hypoglycemia	-Supply may vary, pre-mixed	-Varies	-Pediatrics, variable	
D50W	-Hypoglycemia, hyperkalemia	-Prefilled syringe	-25g in 50mL (0.5g/mL)	-0.5-1 syringe PRN for indication	

MEDICATION	INDICATION	FORMULATION	FINAL CONC	USUAL DOSING	NOTES
Dimenhydrinate **IV or IM**	-Antiemetic	-25-50mg IV/IM q 4-6h prn	-Administer undiluted or dilute with 9mL of NS in a 10mL syringe -5mg/mL	-25-50mg over 2min	-Max dose: 400mg/24h -DO NOT DILUTE IM DOSE
Diphenhydramine **IV or IM**	-Mild cutaneous allergic reaction	-25-50mg IV/IM q 4h prn	-Administer undiluted, or dilute with 9mL of NS in a 10mL syringe -5mg/mL	-25-50mg over 1 min	-Max single dose: 100mg -Max daily dose: 400mg/24h -DO NOT DILUTE IM DOSE
Ephedrine	-Hypotension esp. with bradycardia	-Add 50mg of Ephedrine + remaining vol. NS to equal 10mL total	-5mg/mL	-5-10mg IVP	-May need to repeat q 2-5min -Will also ↑ HR
Epinephrine **IV**	**-Hypotension-mod/severe**	-Add 1mg Epi to 100mL NS mini bag	-10mcg/mL	-10-100mcg q 2-5min prn	-PUSH DOSE EPI
Epinephrine **IV**	**-ACLS/Cardiac Dose**	-Premixed syringe	-1mg/10mL	-1mg IV q 3min	-Cardiac arrest dose **ONLY**
Epinephrine **IM**	**-Anaphylaxis (IM)** ***1st line***	-Epipen x 1 dose OR -Draw up 0.5mL undiluted in a 1mL syringe	-1mg/mL	-0.5mg (0.5mL) IM	-May need to repeat IM dose q 15 min prn -Preferred site: vastus lateralis
Epinephrine **IV**	**-Anaphylaxis (IV)** For ongoing tmt in acute care/critical care setting	-Use pre-filled Epi (ACLS) 1:10,000	-0.1mg/mL (100mcg/mL)	-1mcg/kg IV q 2min prn	-Prepare epi infusion as req. -NOTE: only for acute care/critical care setting -**IM is treatment regimen of choice for most cases of anaphylaxis**
Etomidate **(Amidate)**	-Induction anesthesia	-Draw up 20mg in 10mL syringe (do not dilute) OR -Use pre-filled cartridge	-2mg/mL	-0.2-0.6 mg/kg (adult= 20- 60mg) -usually ½ of pre-filled cartridge	-Lasts approx. 5 min -No analgesic effects -Minimal effect on respiratory/cardiovascular system -Pain on injection
Fentanyl	-Analgesia-opioid	-Administer undiluted	-50mcg/mL	-25-100mcg prn -Titrate to effect	-Caution with >250mcg IVP -Can cause skeletal muscle rigidity w/ rapid injection of larger doses -Antidote: *Naloxone*
Flumazenil	-Reversal-benzo overdose	-Draw up direct (do not dilute) into 5mL syringe	-0.1mg/mL	-0.2mg q 1min prn -Give dose over 15sec	-Prepare for repeat doses *Max total dose*: 2mg -May cause seizures esp. in those on longer term Bz

MEDICATION	INDICATION	FORMULATION	FINAL CONC	USUAL DOSING	NOTES
Furosemide	-Non-potassium sparing diuretic	-Give dose undiluted OR -Dilute with NS to a total volume of 10mL	-10mg/mL	-20-100mg slow IVP over 2min ≥100mg= *see infusion*	-Prep for rapid diuresis -Monitor K^+ & electrolytes -Rapid= risk of ototoxicity/ hypotension
Haloperidol **IV**	-Antipsychotic/ Agitation -Antiemetic	-Dilute with NS/D5W to a total volume of 10mL	-0.5mg/mL	-Antipsychotic 5mg IVP over 1 min -Antiemetic: 0.5-1mg IV q6H	-Monitor respiratory status, blood pressure, and for EPS (extra-pyramidal side effects -tmt diphenhydramine)
Haloperidol **IM** *preferred*	-Antipsychotic/ agitation	-Administer undiluted	-5mg/mL	-5-10mg IM	-Monitor respiratory status, blood pressure, and for EPS (tmt diphenhydramine)
Hydrocortisone	-Corticosteroid	-Administer undiluted once mixed up according to package	-Varied depending on concentration of Vial	-Doses up to 300mg- administer at 100mg/min	-Compatible with NS or D5W
Hydromorphone	-Analgesia-opioid	-Dilute 2mg/mL vial with NS to a total volume of 10mL OR -Dilute 2 x 10mg/mL vial with 100mL NS	-Final concentration: 0.2mg/mL	-Administer dose over 2-3min	**Caution: different concentrations available** -Watch for respiratory depression/decreased blood pressure
Insulin, Regular	-Hyperkalemia -Crush Syndrome	-Administer undiluted	-1u/mL	-10u IVP	-For hyperkalemia: **MUST GIVE D50W FIRST**
Ketamine	-Induction -Analgesia -Dissociative anesthesia	-Add 100mg of Ketamine + remaining vol. NS to equal 10mL total	-10mg/mL	-RSI: 0.5-2mg/kg slow IVP for induction (adult= 70- 200mg) -Pain: 0.1-0.3mg/kg IVP -25-50mg increments for pain/dissociation prn until desired effect	**Caution: different concentrations available** -Prepare for ↑ secretions & consider atropine -When titrating to dissociation, observe for nystagmus
Ketorolac **IV or IM**	-Analgesia-NSAID	-Administer Undiluted	-Vial: 30mg/mL	-15mg/dose IV/IM -Administer dose over 1-2min	-Max dose: 120mg/day
Labetolol	-Hypertension	-Administer Undiluted	-Vial: 5mg/mL	-10-20mg IV over 2min, then 40-80mg q10 min prn	-Max total dose 300mg -Monitor for orthostatic hypotension, bradycardia, bronchospasm
Magnesium Sulfate ***Cardiac Arrest***	-Cardiac arrest, Torsades	-Add 2g Mag + remaining vol. NS to equal 10mL total	-200mg/mL	-Give 2g over 2min (emergency only)	-Refer to infusion for non-urgent dose
Methylprednisolone	-Corticosteroid	-Administer undiluted once mixed up according to package	-Varied depending on concentration of Vial	-IV doses up to and including 125mg to be administered over at least 5min	-Compatible with NS or D5W

MEDICATION	INDICATION	FORMULATION	FINAL CONC	USUAL DOSING	NOTES
Metoclopramide	-Antiemetic	-Administer Undiluted	-Vial: 10mg/mL	-Administer over 1-2 min	-Monitor for EPS-tmt diphenhydramine
Metoprolol	-Antiarrhythmic	-Administer undiluted	-1mg/mL	-2.5-5 mg IV direct every 2-5 minutes	-Maximum of 15 mg over 10-15 minutes
Midazolam ***Anxiety***	-Sedation/amnesia -anxiolysis	-Add 10mg Midazolam + remaining vol. NS to equal 10mL total	-1mg/mL	-0.3mg/kg induction -1-2mg IV prn for amnesia/anxiolysis	**Caution: different concentrations available** -Monitor for hypotension, resp depression
Midazolam ***Seizure***	-Seizure	-2mg/mL OR -10mg/mL	-Draw up 10mg total -5mL 2mg/mL OR -1mL 10mg/mL	-10mg IM/IV	**Caution: different concentrations** -Monitor for hypotension, resp depression
Morphine	-Analgesia-opioid	-Add 10mg Morphine + remaining vol. NS to equal 10mL total	-1mg/mL	-Give 1-3mg increments slow IVP prn	-Prepare for hypotension/ ↓ resp. drive
Naloxone	-Opioid overdose/ Opioid reversal	-Dilute 0.4mg + remaining vol. NS to equal 10mL total OR -Use pre-filled cartridge	-0.04mg/mL OR -1mg/mL (pre-filled cartridge)	-Overdose: 0.4-2 mg IV/IM q 2-3min prn -Reversal post-op: 0.04- 0.08 mg IV q 2-3min prn	-Prepare for repeat doses
Ondansetron	-Antiemetic	-4mg undiluted IVP over 2-5 mins -Dilute 8mg in 50-100 mL of NS or D5W; infuse over 15 minutes.	-4mg/mL OR -0.16mg/mL (50mL) or 0.08mg/mL (100mL)	-4-8mg IV Q6H for nausea	
Phenylephrine	-Hypotension	-10mg in 100mL NS bag -Withdraw 10mL in a syringe for IVP	-100mcg/mL	-100-200mcg IV prn	-Lasts 1-2 min -May cause reflex bradycardia
Propofol	-Anesthesia induction & maintenance	-Draw up 10-20 mL in syringe	-10mg/mL OR -20mg/mL	-1-2mg/kg for induction -May need up to 4mg/kg for pediatric & young healthy pts	**Caution: different concentrations** -Monitor for hypotension
Rocuronium (RSI)	-Paralytic (non-depolarizing)	-Draw up 100mg in 10mL syringe (do not dilute). -May need 2 syringes	-10mg/mL	-0.6-2mg/kg (adult= 50mg-200mg)	-Lasts approx. 30-90min or longer
Scopolamine	-Anti-emetic/ secretions	-Draw up 0.4mg (1mL) into 3mL syringe (do not dilute)	-0.4mg/mL	-0.4mg (1mL) IM/ IV/ SC	-For management of secretions, palliative care

MEDICATION	INDICATION	FORMULATION	FINAL CONC	USUAL DOSING	NOTES
Sodium Bicarbonate	-Many	-Prefilled Syringe	-50mmol of NaHCO3 per syringe	-1-2 syringes PRN	
Succinylcholine (RSI)	-Paralytic (depolarizing)	-Draw up 200mg (10mL) into 10mL syringe (do not dilute)	-20mg/mL	-1-2 mg/kg (adult=50mg-200mg)	-Observe for fasciculation -Lasts approx. 5-10min
Tranexamic Acid (TXA)	-Anti-fibrinolytic (give for significant/ uncontrolled bleed) -MTP	-Draw up 2 x 1g (10mL ea) in 2 x 10mL syringes	-100mg/mL (undiluted)	-2g slow IVP	Rapid infusion may cause hypotension & seizures. 1st dose should be given within first 3 hrs of injury
Vasopressin	-Hypotension	-Add 20 units Vasopressin + remaining vol. NS to equal 10mL total	-2 units/mL	-1-2 units prn	
Vecuronium	-Paralytic (non-depolarizing)	-Reconstitute 10mg (powder) vial with 10mL NS	-1mg/mL	-0.08- 0.1 mg/kg (adult= 5.5-10mg)	-Lasts approx. 60min

IV INFUSIONS: CF AE Flt Revision

MEDICATION	INDICATION	FORMULATION	FINAL CONC	ADMIN	USUAL DOSING	NOTES
Amiodarone (Cordarone)	-Treatment/prophylaxis of VT/VF, SVT **-Cardiac arrest (give undiluted)**	-900mg in 500mL D5W	-1.8mg/mL	mg/min	1) Load: 150-300mg in 100mL D5W, give over 10min, then 2) Maint: 1mg/min x 6h (33mL/h), then 3) 0.5mg/min x 18h (17mL/h)	-Mix in non-PVC bag or glass bottle -0.22 micron filter -Observe for hypotension
Dexmedetomidine (Precedex)	-Sedation	-Add 200mcg (2mL) to 48mL NS for a total of 50mL	-4mcg/mL	mcg/kg/hr	1) Load: 1mcg/kg over 15 min, then 2) Maint: 0.2-1.5mcg/kg/h	-Monitor for hypotension & bradycardia
Diltiazem (Cardizem)	-Ctrl rapid ventricular rate in A-fib/ flutter -Convert PSVT	-125mg(25mL) in 100 mL NS -Total Volume= 125mL	-1mg/mL	mg/hr	1) Load:0.25mg/kg over 2min, then 2) Maint: 5-15mg/h	-Monitor for hypotension
Dobutamine (Dobutrex)	-Hypotension (inotrope)	-500mg in 250mL NS or D5W	-2mg/mL (2000mcg/mL)	mcg/kg/min	2-20mcg/kg/min	-Central line preferred
Dopamine (Intropin)	-Hypotension	-400mg in 250mL D5W (premixed)	-1600mcg/mL	mcg/kg/min	1- 50mcg/kg/min	-Central line preferred -Start at 5-10mcg/kg/min and titrate to effect
Epinephrine (Adrenalin)	-Hypotension (vasopressor)	-1 mg in 100mL NS or D5W OR -1 mg in 250mL OR -High Dose Concentration: 4mg in 250mL	-10mcg/mL OR -4mcg/mL OR -16mcg/mL	mcg/min	2-20mcg/min (0.1-0.5mcg/kg/min)	-Central line preferred esp. in high dose preparations
Esmolol (Brevibloc)	-Beta blocker (control of A-fib/flutter, HTN)	-2,500mg in 250mL (premixed)	-10mg/mL	mcg/kg/min	1) Load: 500mcg/kg over 1min, then 2) Maint: 50mcg/kg/min for 4min If no effect, repeat load	-Central line preferred but not required
Fentanyl	-Analgesic - opioid	-1000mcg in 100mL NS	-10mcg/mL	mcg/h	25-200mcg/h	-Monitor for hypotension, hypoventilation
Furosemide (Lasix)	-Diuretic	-Reconstitute ordered dose to make 1mg/mL	-1mg/mL	mg/h	Doses >20mg should be given on a syringe pump at a max rate of 4mg/min	-Prep for rapid diuresis -Monitor K+ & electrolytes -Rapid= risk of ototoxicity
Heparin	-Anticoagulant	-25,000units in 500mL D5W (premixed)	-50 units/mL	units/h	12-18 units/kg/h	*refer to protocol

MEDICATION	INDICATION	FORMULATION	FINAL CONC	ADMIN	USUAL DOSING	NOTES
Hydromorphone	-Analgesic - opioid	-20mg in 100mL	-200mcg/mL	mg/hr	0.2-0.4mg/h	-Monitor for hypotension/hypoventilation
Hydroxycobalamin	-Cyanide poisoning	-Cyanokit – administer per instructions	-5g in 200mL		-5g over 15-30 mins	
Hypertonic Saline 3%	-Increased ICP	-Premixed in 250mL bag	N/A	mL/h	1) Load: 250mL 3% saline over 10min, then 2) Maint: infuse at 50-100mL/h	-Central line preferred *titrate for serum Na+ of 150-160mEq/L*
Insulin	-Glucose control	-100 Units NPH in 100mL	-1 unit/mL	units/h	1- 20 units/h	*refer to sliding scale/protocol
Isoproterenol	-Cardiac arrest/shock	-1mg in 250mL	-4mcg/ mL	mcg/min	0.5-5mcg/min	-High Dose: 20mcg/min
Ketamine	-Dissociative anesthesia -Analgesia	-100mg in 100mL D5W or NS	-1mg/mL	mg/ kg/h	Anes Induction: 1-2 mg/kg Anesthesia: 0.3-1.8mg/kg/h Pain: 0.03-0.3mg/kg/h	-Caution w/ adults (emergence reaction) -Prepare for ↑ secretions
Labetalol	-Hypertension	-Remove 10mL from 250mL bag of D5W -Add 300mg (60mL) Labetalol	-1mg/mL	mg/min	0.5-2mg/min	-Monitor for orthostatic hypotension, bradycardia, bronchospasm
Levetiracetam (Keppra)	-Anticonvulsant	-1 g in 100mL NS	-10mg/mL	mg/h	500-1000mg diluted in 100mL NS, given over 15-30min	
Lidocaine	-Treatment/ prophylaxis of VF/VT	-1g in 250mL D5W/ NS	-4mg/mL	mg/min	1) Load: 0.5-1.5mg/kg over 2min, then 2) Maint: 1-4 mg/min	
Magnesium Sulfate	-Preeclampsia, A-fib, Bronchospasm	-4g in 100mL D5W or NS	-40mg/mL (0.04g/mL)	g/h	1) Load: 1-2g over 30-60min, then 2) Maint: 0.5-1g/h	-Rapid bolus can cause dysthymia/hypotension
Magnesium Sulfate	-Eclampsia (Pregnancy)	-Add dose to minibag NS	-Varied depending on dose		-4-6g Loading dose over 15-30min, then 1-2g/h	-Monitor for muscle weakness, hypotension
Mannitol 20%	-Osmotic diuretic (increased ICP)	-100g in 500mL (premixed)	-0.2g/mL	g/h	1) Bolus: 1-2g/kg over 30min, then 2) Maint: 1g/kg q 6h over 1h HOLD for Osmo>340mmol or Osm gap >10 OR Na+ >160mmol	Use 0.22 micron IV filter Monitor serum lytes Draw labs 1h b/f repeat dose Monitor BP ↓

MEDICATION	INDICATION	FORMULATION	FINAL CONC	ADMIN	USUAL DOSING	NOTES
Midazolam (Versed)	-Sedation/ amnesia	-100mg in 100mL NS	-1mg/mL	mg/h	1-7mg/h (0.02-0.1mg/kg/hr)	-Monitor Peds for paradoxical reaction (agitation)
Milrinone	-Inotrope - CHF	-20mg in 80mL NS or D5W -Total Volume: 100mL	-200mcg/mL	mcg/kg/min	1) Load: 50mcg/ kg over 10min, then 2) Maint: 0.4-1mcg/kg/min	-Monitor for hypotension
Morphine	-Analgesic - opioid	-100mg in 100mL NS	-1mg/mL	mg/h	1-10mg/h	-Monitor for hypotension & resp. depression
Nitroglycerin	-Unstable angina/MI, CHF w/ acute pulmonary edema, hypertensive emergency	-50mg in 250mL D5W (premixed in glass bottle)	-200mcg/mL	mcg/min	**SCAPE dosing:** Start at 400mcg/min for 2 mins, THEN decrease to 100mcg/min and titrate up for effect **Angina if unresponsive to SL NTG**: 5mcg/min, increase by 5mcg/min q3-5min up to 20mcg/min, THEN increase by 10mcg/min. May later increase by 20mcg/min.	-Monitor for hypotension -Will cause headache -Do NOT administer to pt's taking Viagra/ Revatio (Sildenafil), Cialis, Levitra -Higher doses require PVC admin sets. -Uninterrupted infusions >24h produce tolerance -Caution/contra-indicated in right sided MI, SBP <90, HR >100 or < 50
Nitroprusside	-Hypertensive emergency	-50mg in 250mL D5W	-200mcg/mL	mcg/kg/min	Start at 0.25mcg/kg/min, titrate up by 0.5mcg increments q 2min Target range= 1-3mcg/kg/min Max=10mcg/kg/min	-Do NOT administer to pt's taking Viagra/ Revatio (Sildenafil) -Art line if possible -Monitor for cyanide tox -Protect med from light
Norepinephrine (Levophed) *weight based dosing *mcg/kg/min*	-Hypotension	-4mg in 250mL D5W/NS (regular strength) OR -8mg in 250mL (double strength)	-16mcg/mL OR -32mcg/mL	mcg/kg/min	Start 0.05-0.01mcg/kg/min titrate to desired MAP Range: 0.01-3mcg/kg/min Peds: max 1-2 mcg/kg/min	-Central line preferred esp. in high dose formulation
Norepinephrine (Levophed) *non-weight based *mcg/min*	-Hypotension	-4mg in 250mL D5W/NS (regular strength) OR -8mg in 250mL (double strength	-16mcg/mL OR -32mcg/mL	mcg/min	Start 5-10mcg/min, titrate to desired MAP Range: 2-20mcg/min (max 30mcg/min)	-Central line preferred esp. in high dose formulation

MEDICATION	INDICATION	FORMULATION	FINAL CONC	ADMIN	USUAL DOSING	NOTES
Pantoprazole (Pantoloc/ Protonix)	-GI bleed	-80mg in 100mL D5W or NS	-0.8mg/mL	mg/h	1) Load 80mg over 15min, then 2) Maint: 8mg/h for 72hrs (10mL/h)	
Phenylephrine (Neosynephrine)	-Hypotension	-10mg in 100mL NS or D5W	-100 mcg/mL	mcg/min	Usual: 40-300mcg/min	-High dosing can cause bradycardia
Phenytoin (Dilantin)	-Anti-convulsant	-Dilute dose in 100mL NS	-mg/mL	mg/min	Seizures: 1) Load 15-20mg/kg (Admin no faster than 50mg/min), then 2) Maint: 5mg/kg/day (given over 1h- may be divided in 2 doses) Neuro prophylaxis: 100-200mg q4h	-Use 0.22 micron filter -Monitor for hypotension -Monitor site for extravasation -Flush line well b/f & after -Therapeutic level: 40-80mcmol/L
Procainamide	-VT (with pulse)	-1000mg in 250mL D5W or NS	-4mg/mL	mg/min	1) Load: 17mg/kg IV, then 2) Maint: 2.8mg/kg/h	-Max dose: 20mg/min (300mLh) -Monitor for hypotension, arrhythmia, stop infusion if either develops
Propofol (Diprivan)	-Sedation, anesthesia	-1000mg in 100mL (premix in glass) OR -2000mg in 100mL (premix in glass)	-10mg/mL OR -20mg/mL	mcg/kg/min OR mcg/kg/hr	5-50mcg/kg/min OR 0.3-3mg/kg/hr	**Caution: different concentrations** -Monitor for hypotension -Do not use plastic adaptors change tubing q 12
Rocuronium (Zemuron)	-Paralytic (non-depolarizing)	-500mg in 100mL NS	-5mg/mL	mcg/kg/min	10-12mcg/kg/min (titrate to TOF)	-If possible, test TOF prior to admin Roc
Vasopressin	**Diabetes Insipidus**	-100units in 100mL D5W or NS	-0.1 units/mL	units/h	2.5 units/h, (titrate for ↓ urine output)	-Central line preferred
Vasopressin	**Septic shock**	-100units in 100mL D5W or NS	-0.1 units/mL	units/min	0.01-0.04 units/min	-Usually given as 2nd or 3rd line vasopressor
Vecuronium	-Paralytic (non-depolarizing)	-20 mg in100mL D5W or NS	-200mcg/mL	mcg/kg/min	0.8-1.2mcg/kg/min (titrate to TOF)	

ANTI-INFECTIVE AGENTS CHART: CF AE Flt Revision

MEDICATION (compatible solutions)	DIRECT IV PUSH	INTERMITTENT IV	MEDICATION (compatible solutions)	DIRECT IV PUSH	INTERMITTENT IV
Ampicillin NS, D5W	**MAX(100mg/mL)** 1g in 10mL over 10 min 2g in 20mL over 20 min (*Faster than 100mg/min could cause seizures*)	**MAX (30mg/mL)** 1g in 50mL 2g in 100mL Both over 20-30 min	**Clindamycin** NS, RL, D5W	**NO**	**MAX(18mg/ml)** 300mg in 50mL over 10 min 600mg in 50mL over 20 min 900mg in 100mL over 30 min 1200mg in 100mL over 45 min *Don't exceed 30mg/min*
Azithromycin NS, RL, D5W	**NO**	**MAX (1-2mg/mL)** 250-500mg in 250-500mL over 1-3 hours	**Cloxacillin** NS PREFERRED, D5W	**MAX(100mg/ml)** 1g in 10mL over 2 min 2g in 20mL over 5 min **Recommended Central Line**	**MAX(50mg/mL)** 1g in 50mL & 2g in 100mL both over 30-40 min *use central line/PICC*
Cefazolin NS, RL, D5W, D10W	**MAX (100mg/mL)** 1g in 10mL 2g in 20mL both over 3-5 min	**MAX (20mg/mL)** 1g in 50mL over 15-30 min 2g in 100mL over 30-60 min	**Ertapenem** NS, SWFI	**YES** Safe to administer as 1g in 10mL Slow IVP	**Max(20mg/mL)** Reconstitute 1g with 10mL of compatible solution Further dilute in in 50mL NS Infuse over 30 min
Cefotaxime NS, RL, D5W, D10W	**MAX (100mg/mL)** 1g in 10mL over 3 min 2g in 20mL over 5 min (*If given <1 min may cause arrhythmias)	**MAX(50mg/mL)** 1g in 50mL 2g in 100mL Both over 15-30 min	**Erythromycin** NS, RL	**NO**	**Max(5mg/mL)** 250mg in 100mL 500mg in 250mL All over 30-60 min
Ceftazidime NS, RL, D5W	**MAX(100mg/mL)** 1g in 10mL over 3 min 2g in 20mL over 5 min	**MAX(40mg/mL)** 1g in 50mL 2g in 100mL Both over 15-30min	**Fluconazole** (Y-Site-NS, RL, D5W)	**NO**	**PREMIXED (200mg/100mL)** Max 200mg/hr **Flush before and after with NS**
Ceftriaxone NS, D5W, D10W, *No RL*	**MAX (100mg/mL)** 1g in 10mL over 2 min 2g in 20mL over 5 min	**MAX (40mg/mL)** 1 g in 50mL 2g in 100mL over 15-30 min	**Gentamicin** NS, RL, D5W	**NO**	Dilute dose in 50-200mL D5W or NS Administer over 30 min-2 hours
Cefuroxime NS, D5W	**MAX(100mg/mL)** 750mg in 10mL over 3 min 1500mg in 20mL over 5 min	**MAX(15mg/mL)** 750mg in 50mL 1500mg in 100mL both over 15-60 min	**Imipenem** NS, D5W, D10W	**NO**	**Max(5mg/mL)** 500mg in 100mL over 30 min 750mg-1000mg in 200mL over 40-60 min
Ciprofloxacin (Y-Site- NS, RL, D5W, D10W)	**NO**	**PREMIXED** 60 min or more	**Levofloxacin** (Y-Site-NS, RL, D5W)	**NO**	**PREMIXED** 500mg over 60 min 750mg over 90 min

Created for 1 Cdn Fd Hosp in 2016 by LCol Schmid, Revised in 2020 by Capt Goulet
Revised for CF AE Flt 19 Sept 2021 by Capt Kuipers with LCol Doyle/Maj Wilson

MEDICATION (compatible solutions)	DIRECT IV PUSH	INTERMITTENT IV	MEDICATION (compatible solutions)	DIRECT IV PUSH	INTERMITTENT IV
Linezolid (Y-Site-NS, RL, D5W)	**NO**	**PREMIXED** 600mg in 300mL over 30-120 min	**Piperacillin** NS, RL, D5W	**MAX(180g/mL)** 3g in 10mL over 3 min 4g in 20mL over 5 min	**MAX(80mg/mL)** 3-4g in 50mL over 20-30 min
Meropenem NS, RL, D5W, D10W	**MAX(50mg/mL)** 500mg in 10mL over 5 min 1g in 20mL over 5 min	**MAX(50mg/mL)** Further dilute in a minibag and administer over 15-30 min	**Piperacillin-Tazobactam** NS, D5W, RL	**NO**	**MAX(80mg(piperacillin)/mL)** 2.25g-4.5g in 50-150mL all over 30 min
Metronidazole (Y-Site-NS, RL, D5W)	**NO**	**PREMIXED** 500mg over 20 min Max rate 25mg/min	**Septra** (Sulfamethoxazole/ Trimethoprim) NS, RL, D5W	**NO**	**MAX(1mL in 10mL)-fluid restricted patient** Dilute 1 amp (5mL) in 100-125 mL over 60-90 min *MAX conc. May cause precipitate*
Micafungin NS, D5W	**NO**	**MAX(4mg/mL)** 50-150mg in 100mL over 60 min *Above 1.5mg/mL use central line*	**Tobramycin** NS, RL, D5W	**NO**	Dilute dose into 50-100mL of D5W or NS and administer over 20-60 mins
Penicillin G NS, D5W	**NO**	Dilute dose in 100 mL of compatible solution and give over 30 min-2 hours	**Vancomycin** NS, RL, D5W	**NO**	**MAX Central (10mg/mL) Peripheral (2.5mg/mL-preferred-5mg/mL)** Under 1g over at least 60 min 1-1.5g over at least 90 min Over 1.5g at least 120 min *Do not exceed 10mg/min*

BLOOD PRODUCT CHART: CF AE Flt Revision

BLOOD COMPONENTS	INDICATION	COMPATIBILITY	ADMINISTRATION
RBCs	-Massive Transfusion Protocol -Active Bleeding -Goal to maintain oxygen carrying capacity -Restore Blood Volume -Symptomatic anemia -Will Use Whole Blood First in MTP for trauma...if available	-Must be ABO and Rh compatible -Type & Screen Req if type specific RBCs being used.	-Use Standard Y-Type blood tubing with 170-260 microns filter (Gravity or Pressure Bag) -4 units total or 8 hours total time -Can be given through the Belmont/infusion device (MTP) -Can be given extremely fast but watch for allergic reactions or fluid overload with multiple units -Must replace Calcium due to Citrate in RBC bags
FRESH FROZEN PLASMA	-Massive Transfusion Protocol -Active Bleeding -Goal for coagulation factors replacement -Restore Blood Volume	-Must be ABO compatible -Type & Screen Req if type specific plasma being used -Rh compatibility NOT Req	-Use Standard Y-Type blood tubing with 170-260 microns filter (Gravity or Pressure Bag) -4 units total or 8 hours total time -Can be given through the Belmont/infusion device (MTP) -Can be given extremely fast but watch for allergic reactions or fluid overload with multiple units
FIBRINOGEN CONCENTRATE (RiaSTAP & FIBRYGA)	-Massive Transfusion Protocol -Active Bleeding -Goal for fibrinogen replacement	-Type & Screen NOT Req -Dose/Frequency based on the extent of bleeding, lab results and clinical condition of the patient	-Does NOT require blood tubing but does require its own line -Can be hung as a primary line or given IVP -Reconstitute according to manufacture insert, normally with 50mL SWFI -use within 8 hours after reconstitution -Current dose is 3-4g for first Trauma -RiaSTAP ~ 1g/vial -Fibryga ~ 2g/vial

BLOOD COMPONENTS	INDICATION	COMPATIBILITY	ADMINISTRATION
CRYOPRECIPITATE (Not available in all deployments)	-Massive Transfusion Protocol -Active bleeding -Goal for fibrinogen replacement + Factors VIII, XIII, vWF	-ABO compatible NOT Req -Rh compatibility NOT Req	-Blood tubing required -Transfuse as quickly as possible
ALBUMIN (Not available on all deployments)	-Fluid resuscitation in critically ill patients (except those with TBI) who are unresponsive to crystalloids or in whom they are contra-indicated. -Fluid resuscitation in patients with advanced liver cirrhosis -Contra-indicated in TBI -Available in 5% and 25% concentration	Type and Screen NOT Req	-Use vented tubing without a blood filter -Does NOT go through Belmont -MAX rate (25%) is 120mL/hr, could cause fluid overload -Max rate (5%) is 300mL/hr -Use within 4 hours of being spiked
PLATLETS (Not available on all deployments)	-Massive Transfusion Protocol -Active bleeding -Goal for replacement in acute hemorrhage OR prevention of spontaneous bleeding in pts with low plt count (<10000/L)	-ABO and Rh identical preferred but NOT required -Type & Screen Req -Rh must match women of child bearing potential, if not then anti-D Immunoglobulin should be considered	-Separate Blood Tubing Req -Does NOT go through Belmont -Can be given rapidly, do not use an infusion device -Change tubing every 4 units or after 8 hours

References:

1) The Ottawa Hospital (TOH), Parenteral Drug Therapy Manual, 37th Edition, 2016
2) Queensway Carleton Hospital, Pharmacy Department-Information Sheet, 2018
3) The Civic Campus ICU Critical Care, Drug Infusion Preparation Table, Ottawa, ON, 2017
4) Lexicomp, accessed February 2018
5) Bloody Easy 4, Blood Transfusions, Blood Alternatives and Transfusion Reactions, 4th Edition, 2016
6) Bloody Easy, Blood Administration, Version 2, November 2015
7) CC-75-005 Blood Transfusions: Administration of Blood, Blood Components and Blood Products, Appendix 3, Capital District Health Authority, 2015
8) Advanced Cardiovascular Life Support Provider Manual, 2016
9) Joint Trauma System Clinical Practice Guidelines – Tactical Combat Casualty Care Guidelines, 2020